Day 29: Dimensionality Reduction & PCA Core

MATHEMATICAL THEORY, COMPREHENSIVE NUMERICAL SOLUTION & COVARIANCE ALGEBRA

1. Theoretical Foundations of Principal Component Analysis (PCA)

Definition (The Principal Component Matrix): Principal Component Analysis (PCA) is an advanced orthogonal linear transformation technique utilized in unsupervised machine learning to perform dimensionality reduction. It systematically maps high-dimensional vector spaces into lower-dimensional subspaces by maximizing the variance along new orthogonal trajectories, effectively eliminating structural multi-collinearity without losing critical underlying information parameters.

> *PCA ek data compression tool hai jo bohot saare columns (High Dimensions) ko kam se kam columns (Low Dimensions) mein badal deta hai. Yeh data ko delete nahi karta, balki unke aapas ke rishte ko nichodkar sabse zyada information wale naye VIP columns banata hai जिन्हें Principal Components kehte hain.*

The Core Historical Practical Origin: Formulated fundamentally by Karl Pearson in 1901 and expanded into mathematical systems using matrix components by Harold Hotelling in 1933, PCA resolves the "Curse of Dimensionality." When data engines face an excessive number of features, statistical models degrade due to continuous multi-dimensional dispersion. PCA isolates the definitive directional orientation vectors, allowing real-world operations to run complex predictive algorithms over tight, high-density matrix architectures.

> *Iska pehla concept Karl Pearson ne 1901 mein geometry aur variance ke liye diya, aur 1933 mein Harold Hotelling ne isme Eigenvalues aur Matrix Math jodkar ise poora kiya. Iska fayda yeh hai ki jab aapke paas 50 columns ho jate hain toh graph banana ya trend pakadna namumkin ho jata hai, PCA unhe 2-3 columns mein badal kar visualization aasan kar deta hai.*

2. The Step-by-Step Mathematical Framework

To execute a complete dimensionality reduction process, the mathematical engine proceeds sequentially through structural matrix stages:

Stage I: Mean Centering (Translation to Vector Origin): The mean of every discrete independent vector feature is computed and subtracted from each constituent parameter score, shifting the collective geometric center directly to the origin coordinate system (0,0).

> *Sabse pehle har column ka average (mean) nikal kar use asli data se minus kiya jata hai, jisse poora graph uthakar zero point (0,0) par aa jata hai taaki aage ki calculus aur matrix math seedhi ho sake.*

Stage II: Covariance Matrix Construction: The system builds a symmetric covariance matrix to track how multiple independent metrics scale dynamically alongside one another across all active coordinates.

> *Iske baad ek Covariance Matrix banaya jata hai jo har ek variable ka dusre variable ke sath aapas ka rishta (correlation) aur badlav ki raftar ko map karta hai.*

Stage III: Eigenvalue & Eigenvector Decomposition: Eigenvectors define the exact geometric direction vectors (Principal Components) along which variance scales to its maximum. Eigenvalues quantify the structural magnitude and strength contained inside those specific directional coordinates.

> *Linear Algebra lagakar Eigenvalues aur Eigenvectors nikale jaate hain. Eigenvector batata hai rasta (Direction) jahan data sabse lamba phaila hai, aur Eigenvalue batati hai us raste ki taqat (Variance).*

3. Comprehensive Numerical Scenario & Mathematical Solution

To evaluate a production scenario within commercial retail models (Goda Gold Dubai), we analyze a two-dimensional matrix containing two distinct operational customer records:

> **Initial Structural Data Logs:**
> 1. Customer A: Income (X) = 4 Lakhs, Gold Purchase (Y) = 2 Tolas
> 2. Customer B: Income (X) = 8 Lakhs, Gold Purchase (Y) = 4 Tolas

Objective Statement: Apply full analytical Principal Component Analysis decomposition to compress this two-dimensional dataset into a single primary dimension vector (PC1).

> *Hamare paas do features hain: Income (X) aur Purchase (Y). Hume PCA ka pura math lagakar in dono columns ko khatam karna hai aur sirf ek VIP column (PC1) banana hai.*

Step-by-Step Analytical Execution Process

Step 1: Mean Centering Computation

Mean of Vector X: Mean(X) = (4 + 8) / 2 = 6

Mean of Vector Y: Mean(Y) = (2 + 4) / 2 = 3

Centered Customer A Point = $\begin{vmatrix} -2 \\ -1 \end{vmatrix}$

Centered Customer B Point = $\begin{vmatrix} 2 \\ 1 \end{vmatrix}$

> *Dono features ka mean nikala (6 aur 3). Phir use minus kiya toh centered points mile real vertical brackets matrix form mein.*

Step 2: Covariance Matrix Construction

Variance(X) = [(-2)^2 + (2)^2] / 2 = (4 + 4) / 2 = 4

Variance(Y) = [(-1)^2 + (1)^2] / 2 = (1 + 1) / 2 = 1

Covariance(X,Y) = [(-2 * -1) + (2 * 1)] / 2 = (2 + 2) / 2 = 2

Assembled Covariance Matrix C = $\begin{vmatrix} 4 & 2 \\ 2 & 1 \end{vmatrix}$

> *Data se Variance aur Covariance nikala, jisse hamara symmetric matrix bana ekdum textbook bracket style mein.*

Step 3: Finding Eigenvalues via Characteristic Equation

Determinant(C - λI) = 0

$$| C - \lambda I | = \begin{vmatrix} 4 - \lambda & 2 \\ 2 & 1 - \lambda \end{vmatrix} = 0$$

(4 - λ)(1 - λ) - (2 * 2) = 0

4 - 4λ - λ + λ^2 - 4 = 0

λ^2 - 5λ = 0 → λ(λ - 5) = 0

Derived Characteristic Roots: $\lambda_1 = 5$, $\lambda_2 = 0$

> *Determinant equation ko solve kiya toh do values mili: $\lambda_1 = 5$ (sabse badi taqat - PC1) aur $\lambda_2 = 0$ (isey drop kar denge).*

Step 4: Extracting the Primary Eigenvector (PC1 Direction)

Using primary root $\lambda_1 = 5$ into systemic equation $(C - \lambda I)v = 0$:

$$\begin{vmatrix} 4 - 5 & 2 \\ 2 & 1 - 5 \end{vmatrix} \begin{vmatrix} v1 \\ v2 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix}$$

$$\begin{vmatrix} -1 & 2 \\ 2 & -4 \end{vmatrix} \begin{vmatrix} v1 \\ v2 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix}$$

Equation: $-1(v1) + 2(v2) = 0 \rightarrow v1 = 2(v2)$

Base Eigenvector Vector = $\begin{vmatrix} 2 \\ 1 \end{vmatrix}$

Length calculation: $\mathrm{Sqrt}(2^2 + 1^2) = \mathrm{Sqrt}(5)$

Normalized Unit Eigenvector Vector (PC1) = $\begin{vmatrix} 2 / \mathrm{Sqrt}(5) \\ 1 / \mathrm{Sqrt}(5) \end{vmatrix} = \begin{vmatrix} 0.894 \\ 0.447 \end{vmatrix}$

> *Bade eigenvalue 5 ko matrix mein rakhkar equation nikali, jisse vertical vector aur uska normalized unit vector mila. Yeh hamara asli optimized rasta hai.*

Step 5: Vector Coordinate Projection

Multiplying the continuous centered data points by the normalized unit eigenvector matrix (Dot Product):

$$\textbf{Projected Coordinate A (PC1 Value)} = \begin{vmatrix} -2 & -1 \end{vmatrix} * \begin{vmatrix} 0.894 \\ 0.447 \end{vmatrix}$$

= (-2 * 0.894) + (-1 * 0.447) = -1.788 - 0.447 = **-2.235**

$$\textbf{Projected Coordinate B (PC1 Value)} = \begin{vmatrix} 2 & 1 \end{vmatrix} * \begin{vmatrix} 0.894 \\ 0.447 \end{vmatrix}$$

= (2 * 0.894) + (1 * 0.447) = 1.788 + 0.447 = **2.235**

> *Ab centered data ko is vertical optimized vector se multiply kiya toh dono purane coordinates badal kar sirf ek VIP column ki value ban gaye.*

Final Systemic Dimensions Reduced Log

The matrix transformation scales down two independent variable parameters into a single optimized array:

Customer	Income Vector (Old X)	Purchase Vector (Old Y)	Projected PC1 Vector (New Variable)
Customer A	4	2	**-2.235**
Customer B	8	4	**2.235**

Formal Interpretation: The data pipeline completely collapses two original features into one synthetic variable (PC1) containing 100% of the variance distribution system. The secondary component evaluates exactly to zero information score, validating complete dimensionality reduction without parameter loss.

> *Extraction: Dono columns khatam ho chuke hain. Hamara poora table ab sirf ek single column (PC1) par chal raha hai jo purane dono data points ka 100% impact andar hold karta hai. Optimization done!*